How does the implementation of solar systems impact the development of Central Africa?

Assumptions

When I began this project, I expected to determine that the implementation of solar systems had a positive effect on local socioeconomic conditions in Central Africa. However, I soon found that its effect on socioeconomic conditions was more complex and confounded than a simple scalar boost.

Definitions

Photovoltaics: Materials that convert solar energy into electrical energy

Central Africa: A subregion of the African continent centered primarily around the Congo Basin

Development: Broadly, improvement in health, education, and income, often benchmarked by the inequality-adjusted human development index (IHDI); the sources here measure narrower indicators such as income, energy costs, and health

Solar Resource: The total amount of electromagnetic radiation emitted by the sun that reaches a specific geographic area on Earth

Perspectives

Falchetta focuses on the feasibility and economic potential of solar-based irrigation systems in Sub-Saharan Africa. It found that "irrigation could support smallholder farmers with stabilizing crop yields, increasing incomes, and achieving food security" while "a key barrier to irrigation uptake is inadequate solar electricity … [we devise a] modeling framework to show that over one third of unmet crop water requirements … could be supplied with standalone solar photovoltaic (PV) irrigation systems that can be paid back by farmers within 20 years" (Falchetta et al.).

Maqbool and Akubo's results support this, finding that "predicted that [socioeconomic factors] are the major challenges in hindering the overall sustainability in Africa rather than technical complexities … [a similar outcome] also validated by the Case study of ITC centre powered project of solar energy" (Maqbool and Akubo).

Wassie validates these results with evidence from southern Ethiopia which suggested that "a solar-electrified rural household could save the consumption of 43.68 L of kerosene and emission of 107 kg CO_2 per year compared to a non-electrified one … [which could] allow a rural household to save between US$ 65 and $75 per year from avoided energy costs … reduce health damage from kerosene lamps; and allow micro-businesses to generate more income" (Wassie and Adaramola).

Final Conclusion

My understanding of this issue has grown tremendously through my research. At first, I thought photovoltaics were a decent way to grant Africa, especially Central Africa, electrical energy. However, I have soon come to realize that solar power can be used for a myriad of things, such as solar irrigation systems Falchetta described. I also did not know that equatorial distance and local climate had such a significant impact on the efficacy of photovoltaic systems; I assumed they were OK everywhere since Germany, a rather northern country, had so many of them. Most of all, I now believe that socioeconomic barriers, more than technology, may be what keeps Central Africa from adopting solar systems (Maqbool and Akubo; Fraser).

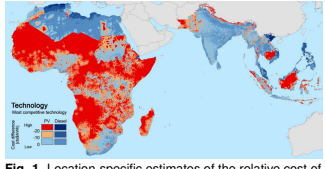


Fig. 1. Location-specific estimates of the relative cost of solar PV versus diesel-based minigrids in the adverse PV conditions scenario. Szábó et al., fig. 4.

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Synthesis

As a whole, these studies suggest a major pattern: solar systems can raise incomes and cut energy costs (Falchetta et al.; Wassie and Adaramola), but whether they are adopted at all may depend more on socioeconomic conditions than on technology (Maqbool and Akubo).